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AI Inference Optimization Techniques (2025-2026)

research

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Executive Summary

AI inference optimization has become critical as LLM deployments scale. This research covers the major techniques driving 2-24x performance improvements: speculative decoding, continuous batching, KV cache optimization, quantization, and specialized inference frameworks.

1. Speculative Decoding

How It Works

Speculative decoding accelerates LLM inference by using a smaller "draft" model to predict multiple tokens ahead, then verifying them in parallel with the larger "target" model. This exploits the fact that verification is much cheaper than generation.

Key mechanisms:

- Draft model generates candidate token sequences (trees or chains)
- Target model verifies candidates in a single forward pass
- Accepted tokens are used; rejected ones trigger regeneration
- **Output is mathematically identical** to standard decoding

Performance Benchmarks

Method	Speedup	Notes
Standard speculative decoding	2-3x	Google's original paper (translation/sum
EAGLE-3	2-6x	Lightweight draft head, 2-5% of target m
SpecEE (Early Exiting)	2.25-2.43x	Tested on Llama2, both cloud and PC sce
High-throughput (batch 256)	2.37x	No architectural changes needed

EAGLE-3: State of the Art

EAGLE-3 (NeurIPS 2025) represents the current best approach:

- **Architecture:** Attaches 1-2 transformer layers as a "draft head" (2-5% of target model size)
- **Innovation:** Uses fusion of low-, mid-, and high-level semantic features
- **Training:** Training-time testing simulates inference

- **Framework support:** vLLM v0.8.5+, SGLang via SpecForge

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Real-World Adoption

- **Google Search:** AI Overviews uses speculative decoding for faster responses
- **vLLM:** Native EAGLE-1/EAGLE-3 support since v0.8.5
- **SGLang:** SpecForge training framework for production deployment

Limiting Factors

- Acceptance rate (α) typically 0.6-0.8 in practice, not near-perfect
- Domain mismatch between draft and target models reduces effectiveness
- Task-specific tuning often needed for optimal results

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2. Continuous Batching

How It Works

Continuous batching (also called iteration-level or dynamic batching) processes requests dynamically rather than as fixed batches:

- New requests join the batch at any generation step
- Completed requests exit immediately
- KV caching avoids recomputing past tokens
- Chunked prefill handles variable-length prompts

Core principle: Instead of waiting for all requests to complete, continuously add/remove requests each token generation step.

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Performance Benchmarks

Scenario	Throughput Improvement
Moderate length variation	2-3x
High length variation	4-8x
vLLM vs baseline	Up to 23x
vLLM vs TGI (high concurrency)	24x
Near-optimal hardware utilization	Achievable

Key Metrics (vLLM benchmarks)

- 4,741 tokens/second at 100 concurrent requests
- Consistent scaling up to batch size 64
- Diminishing returns beyond batch 64

Framework Support

All major frameworks support continuous batching:

- **vLLM:** Core feature
- **SGLang:** Built-in
- **TensorRT-LLM:** "In-flight batching"
- **LMDeploy:** "Persistent batching"
- **Hugging Face TGI:** Supported

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3. KV Cache Optimization (PagedAttention)

The Problem

KV (key-value) cache stores attention states for all generated tokens. Traditional approaches:

- Pre-allocate contiguous memory per request
- 60-80% memory wasted due to fragmentation
- Memory grows quadratically with sequence length

PagedAttention Solution

Inspired by OS virtual memory paging:

- Divides KV cache into fixed-size blocks (default: 16 tokens)
- Blocks stored non-contiguously in memory
- Virtual-to-physical block mapping via block tables
- Near-zero memory waste (<4%)

Performance Impact

Metric	Improvement
Memory waste reduction	From 60-80% to <4%
Throughput vs FasterTransformer	2-4x
Throughput vs early TGI	2.2-3.5x

Metric	Improvement
Memory efficiency	Enables 2-3x larger batch sizes

Advanced Techniques (2025)

- **LMCache:** Hierarchical KV caching (GPU → CPU → network)
- **Prefix caching:** Reuse common prompt prefixes
- **KV compression:** FP8/INT8 KV cache for 2-3x memory savings
- **Automatic prefix caching:** vLLM's built-in feature

Configuration Tips

- Increase `gpu_memory_utilization` for more KV cache space
- Use `tensor_parallel_size` to distribute across GPUs
- Enable prefix caching for repetitive prompts

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4. Quantization

Overview

Quantization reduces model precision from FP16/BF16 to lower bit-widths, trading minimal accuracy for significant memory and speed gains.

Quantization Levels Comparison

Format	Memory (7B model)	Quality Impact	Best Use Case
FP16/BF16	~14 GB	Baseline	Maximum quality
FP8	~7 GB	<1% degradation	Production inference
INT8	~7 GB	~2% degradation	Balanced deployment
INT4	~3.5 GB	8-10% degradation	Edge/resource-constrained
NVFP4	~4 GB	<1% (on Blackwell)	Next-gen GPUs

FP8 Quantization (State of the Art)

- **Hardware:** Requires Hopper (H100) or Ada Lovelace GPUs
- **Speedup:** 2x faster than FP16 with proper kernels
- **Memory:** 7 GB vs 16 GB for 7B model
- **Quality:** Higher dynamic range than INT8, better accuracy

Benchmark (LLaMA-v2-7B on H100):

- 2.3x inference speedup vs FP16
- Batch size 16, latency <500ms
- Input length 1024, output length 128

NVFP4: Next Generation (Blackwell)

- 3.5x memory reduction vs FP16
- 1.8x reduction vs FP8
- <1% accuracy degradation on LiveCodeBench, MMLU-PRO

Quantization Method Rankings

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1. **FP8:** Best for batch ≥16, optimal performance/accuracy
2. **Q5_K_M / GPTQ-INT8:** Best trade-off for most domains
3. **AWQ:** Generally better than GPTQ for weight-only
4. **INT4 (GPTQ):** Use cautiously, significant accuracy loss on small models

Task-Specific Impact

- **Most affected:** Coding, STEM tasks
- **Least affected:** General conversation
- **Recommendation:** 70B+ models can maintain quality at 4-bit; smaller models need 8-bit

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5. Inference Frameworks Comparison

Framework Overview

Framework	Best For	Setup Time	Key Feature
vLLM	High-throughput production	1-2 days	PagedAttention
SGLang	Complex agents/RAG	1-2 days	RadixAttention
TensorRT-LLM	Max single-user perf	1-2 weeks	NVIDIA optimization
llama.cpp	Edge/portability	Hours	CPU-first, any hardware
TGI	HuggingFace ecosystem	Hours	Long context, prefix cac

Performance Benchmarks

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vLLM:

- 14-24x throughput vs HuggingFace Transformers
- 120-160 requests/second
- 50-80ms TTFT (time to first token)
- 4,741 tokens/second at 100 concurrent

SGLang:

- Up to 5x higher throughput in multi-call workloads
- Up to 3.1x higher throughput than vLLM on Llama-70B
- Most stable per-token latency (4-21ms)

TensorRT-LLM:

- Best single-request throughput
- 35-50ms TTFT at low concurrency
- Outperforms all on B200 GPUs
- Requires most engineering effort

llama.cpp:

- Extreme portability (laptops, phones, servers)
- No external dependencies
- 2-bit to 8-bit quantization support
- CPU-optimized

Recommendations by Use Case

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Use Case	Recommended Framework
Interactive apps, high concurrency	vLLM
Agent chains, RAG systems	SGLang
Maximum perf, NVIDIA hardware	TensorRT-LLM
Edge devices, single user	llama.cpp
HuggingFace models, long chats	TGI v3

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6. Mixture of Experts (MoE)

How It Works

MoE architectures activate only a subset of parameters per inference:

- Total parameters: Can be trillions
- Active per token: Typically 5-10%
- Router network selects relevant "experts"
- Enables massive models with manageable compute

Efficiency Gains

Metric	Improvement
Compute per inference	90-95% reduction
Training efficiency	2-7x faster

Metric	Improvement
Power consumption	Up to 50% reduction
Memory per inference	Sub-linear growth

Notable MoE Models (2025-2026)

Model	Total Params	Active Params	Context
DeepSeek R1	671B	37B	Standard
Gemini 1.5	~1T	150-200B	1M tokens
Kimi K2	~1T	32B	Long context
Meta sMLP	Variable	Sparse	3-4x memory reduction

Key Research Advances (2025)

- **Super Experts:** Critical subset of experts that disproportionately affect output
- **MaxScore routing:** Formulates routing as constrained optimization
- **MegaScale-Infer:** Disaggregated expert parallelism for scale
- **NetMoE:** Dynamic sample placement for training acceleration
- **Comet:** Fine-grained computation-communication overlap

Production Considerations

- Expert load balancing crucial for efficiency
- Token dropping can occur under capacity constraints
- Dynamic expert pruning for on-device deployment

- Mixed-precision quantization per expert

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7. FlashAttention

The Memory Problem

Standard attention: $O(N^2)$ memory complexity where N = sequence length

- 2K tokens: Manageable
- 128K tokens: Prohibitive
- 1M tokens: Impossible without optimization

FlashAttention Solution

- Fuses attention operations into single kernel
- Processes data in blocks to maximize GPU cache usage
- **Memory complexity:** O(N) - linear instead of quadratic
- **No accuracy loss:** Mathematically identical output

FlashAttention Version Comparison

Version	GPU	FP16 TFLOPS	Utilization	Key Features
FA-1	A100	~300	~50%	Basic fusion
FA-2	A100/H100	~400	~35%	Improved kernels
FA-3	H100	840	85%	Warp specialization, FP8

Version	GPU	FP16 TFLOPS	Utilization	Key Features
FA-4	Blackwell	TBD	Higher	Blackwell-specific

FlashAttention-3 Performance (H100)

- **BF16:** Up to 840 TFLOPs/s (85% utilization)
- **FP8:** Up to 1.3 PFLOPs/s
- **Speedup vs FA-2:** 1.5-2.0x (FP16), even higher for FP8
- **Memory savings:** 10x at 2K sequence, 20x at 4K sequence

FlashAttention-3 Techniques

1. **Warp specialization:** Overlaps compute and data movement
2. **Pipelined kernel fusion:** Interleaves matmul and softmax
3. **Block quantization:** Hardware FP8 support
4. **Asynchronous TMA:** Tensor Memory Accelerator usage

Context Length Impact

FlashAttention enabled context length explosion:

- 2019: 2-4K (GPT-3, OPT)
- 2023: 128K (GPT-4)
- 2024+: 1M+ (Llama 3, Gemini)

Requirements

- FlashAttention-3: H100/H800, CUDA 12.3+ (12.8 recommended)
- Blackwell GPUs get FA-4 with additional optimizations

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8. Model Serving Platforms

Platform Comparison

Platform	Developer	Strengths	Best For
vLLM	UC Berkeley	PagedAttention, throughput	General productio
TGI	HuggingFace	Ecosystem, long context	HF model users
Triton	NVIDIA	Multi-model, enterprise	Complex pipelines
RayLLM/Anyscale	Anyscale	Auto-scaling, K8s native	Cloud-native deplo

TGI v3 (2025)

- 3x more tokens processed vs previous
- Up to 13x faster on long prompts with prefix caching
- Multi-hardware: NVIDIA, AMD, Intel, Gaudi, Inferentia
- Production-ready with Kubernetes auto-scaling

NVIDIA Triton

- Framework agnostic: PyTorch, TensorFlow, ONNX
- Model ensembles for chaining

- Multi-model serving on single server
- Enterprise-grade monitoring and management

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Anyscale/RayLLM

- Built on Ray Serve
- OpenAI-compatible API
- Auto-scaling across multi-GPU/multi-node
- Private endpoints in your cloud

vLLM Production Stack (2025)

The llm-d project launched by Red Hat, Google Cloud, IBM, NVIDIA, CoreWeave:

- Kubernetes-native distributed serving
- Enterprise support via Red Hat AI Inference Server
- Reference architecture for scale deployments

Companies Using vLLM in Production

- **Amazon:** Rufus
- **LinkedIn:** AI features
- **Meta, Mistral AI, Cohere, IBM:** Core inference
- **Roblox:** Gaming AI

Key Takeaways

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Optimization Priority Order

1. **PagedAttention/Continuous Batching:** 2-24x throughput (framework choice)
2. **FlashAttention:** Enable long context, reduce memory
3. **Quantization (FP8):** 2x speed, 50% memory on H100+
4. **Speculative Decoding:** Additional 2-3x on top of above
5. **MoE architecture:** For new model training

Framework Selection Guide

```
High concurrency + throughput → vLLM
Agent workflows + RAG → SGLang
NVIDIA + max performance → TensorRT-LLM
Edge/portability → llama.cpp
HuggingFace ecosystem → TGI
Enterprise multi-model → Triton
```

Hardware Recommendations

- **Production (H100):** FP8 quantization, FA-3, vLLM/SGLang
- **Edge (RTX 40xx):** llama.cpp, INT4/INT8
- **Next-gen (Blackwell):** NVFP4, FA-4, TensorRT-LLM

Sources

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- FlashAttention-3 Blog
- Cerebrium Framework Benchmarks
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